Final Report

Justin Segrest, Tyler Pollard, Christian Haskell

Introduction

Sleep health is recognized as a cornerstone of overall physiological and psychological well-being. According to global health estimates, nearly 45% of the population suffers from sleep-related issues that compromise quality of life and long-term health outcomes. Despite this, clinical diagnosis often requires extensive and expensive testing. This project explores the potential of using readily available lifestyle and health metrics to identify and categorize sleep disorders.

By leveraging the Sleep Health and Lifestyle Dataset from Kaggle, this research aims to bridge the gap between daily behavioral data and clinical sleep categories: None (Normal), Insomnia, and Sleep Apnea.

Research and Objectives

The core objective of this project is to move from raw data to actionable insight by pursuing two distinct analytical paths:

- **Direction A: Prediction**
  - *Question:* Can a person’s sleep category be accurately classified using only non-invasive metrics such as BMI, stress levels, daily steps, and blood pressure?
  - *Goal:* To develop a machine learning pipeline that maximizes predictive performance and reliability.
- **Direction B: Inference**
  - *Question:* Which specific lifestyle factors act as the most significant "red flags" for sleep disorders?
  - *Goal:* To identify the strongest predictors, such as BMI or Stress Level, and interpret their statistical relationship with the health outcomes.

Methodology Approach

To ensure statistical rigor, we decided to utilize Logistic Regression for clinical interpretability, K-Nearest Neighbors (KNN) to analyze health profiles, and Random Forest for highly-dimensional predictive power. These three methods, supported by a systematic comparison and a paired t-test, ensures that our models are not only accurate, but significant.

Dataset Overview

The dataset consists of 374 individual observations across 13 health and lifestyle variables. The study covers ages from 27 to 59, gender, and profession, and gives a multidimensional dataset for analysis.

Key Outcome Variable: The target variable, Sleep Disorder, is categorical with three levels:

1. None: Individuals with no diagnosed sleep disorder (58.5% of the sample).
2. Sleep Apnea: Individuals experiencing breathing interruptions during sleep (20.9%).
3. Insomnia: Individuals with difficulty falling or staying asleep (20.6%).

While the "None" category represents the majority, the nearly equal split between Sleep Apnea and Insomnia allows for balanced multi-class classification, though it does necessitate the use of stratified K-Fold cross-validation to maintain these proportions during model training.

Data Processing and Descriptive Analysis

Before applying machine learning algorithms, we used various methods in Python to turn Kaggle dataset into a model-ready dataframe. This phase focused on data integrity, feature engineering, and class balancing.

Data Cleaning and Feature Engineering

The raw dataset contained several structural challenges that required manual intervention:

- Handling Implicit Labels: The target variable, **Sleep Disorder**, contained null values for those with a disorder. Rather than dropping these rows, which would've removed half of the dataset, we relabeled them, encoding all "NaN" values as "None." This preserved the 219 healthy control cases necessary for a robust classification.
- Blood Pressure Decomposition: In its raw format, **Blood Pressure** was stored as a string (e.g., "126/83"). Because our machine learning models require numeric inputs, we parsed the strings and split them into two distinct features: **Systolic** and **Diastolic** pressure.
- Ordinal vs. One-Hot Encoding: Categorical variables were handled based on their nature. **BMI Category** was mapped ordinally (Normal: 0, Overweight: 1, Obese: 2) to preserve the physical progression of weight, while **Gender** and **Occupation** were transformed via One-Hot Encoding to avoid the implication of a hierarchy within professions.

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	1	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	NaN
1	2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
2	3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
...	...	...	...	...	...	...	...	...	...	...	...	...	...
369	370	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
370	371	Female	59	Nurse	8.0	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
371	372	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
372	373	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
373	374	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea

Fig 1: Before Data Cleaning was Performed

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Heart Rate	Daily Steps	Sleep Disorder	Systolic	Diastolic
0	1	Male	27	Software Engineer	6.1	6	42	6	Overweight	77	4200	None	126	83
1	2	Male	28	Doctor	6.2	6	60	8	Normal	75	10000	None	125	80
2	3	Male	28	Doctor	6.2	6	60	8	Normal	75	10000	None	125	80
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	85	3000	Sleep Apnea	140	90
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	85	3000	Sleep Apnea	140	90
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
369	370	Female	59	Nurse	8.1	9	75	3	Overweight	68	7000	Sleep Apnea	140	95
370	371	Female	59	Nurse	8.0	9	75	3	Overweight	68	7000	Sleep Apnea	140	95
371	372	Female	59	Nurse	8.1	9	75	3	Overweight	68	7000	Sleep Apnea	140	95
372	373	Female	59	Nurse	8.1	9	75	3	Overweight	68	7000	Sleep Apnea	140	95
373	374	Female	59	Nurse	8.1	9	75	3	Overweight	68	7000	Sleep Apnea	140	95

Fig 2: After Data Cleaning was Performed

Statistical Scaling for Distance-Based Logic

A crucial step for our K-Nearest Neighbors (KNN) model was the application of StandardScaler. Because our KNN model relies on Euclidean distance, features with large numeric ranges (like Daily Steps, which ranges from 3,000 to 10,000) would disproportionately influence the model compared to features with small ranges (such as BMI). By standardizing the features to a mean of 0 and a standard deviation of 1, we ensured a "level playing field" for the health metrics.

Descriptive Summary and Class Balance

The final dataset contains 374 observations. As shown in the descriptive analysis, the dataset had a slight class imbalance:

- None: 58.5%
- Sleep Apnea: 20.9%
- Insomnia: 20.6%

To mitigate this imbalance and ensure certain models did not simply "guess" the majority class, we employed an 80/20 Train-Test split by stratification. This ensures that the ratio of disorders remains consistent across both the training and testing sets, providing a more reliable measure of performance for the minority classes (Apnea and Insomnia).

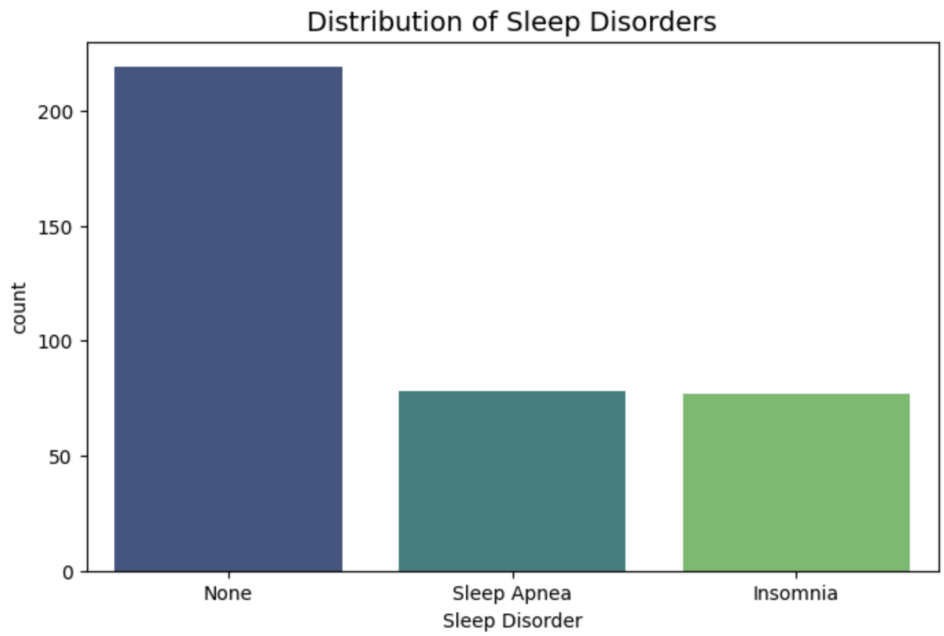

Fig 3: The Distribution of Sleep Disorders in the dataset.

Exploratory Data Analysis (EDA) Highlights

The objective of the EDA phase was to move beyond the simple summary statistics and uncover the hidden structures within the health metrics that drive sleep outcomes. Our analysis revealed three critical relationships that helped with our modeling strategy.

The Correlation Landscape: Identifying Redundancy

To understand the connectivity of the health metrics, a correlation heatmap was generated. The most significant finding was a nearly perfect negative correlation (r = -0.90) between Stress Level and Quality of Sleep. Additionally, Sleep Duration showed a strong positive correlation with Quality of Sleep (r = 0.88).

- Takeaway: This indicates a high degree of multicollinearity in the dataset. While these variables are powerful predictors, including both in a linear model like Logistic Regression can make the coefficients less stable. This finding justified our use of attempting a Random Forest model, an ensemble method that handles redundant, highly-correlated features more effectively than standard linear models.

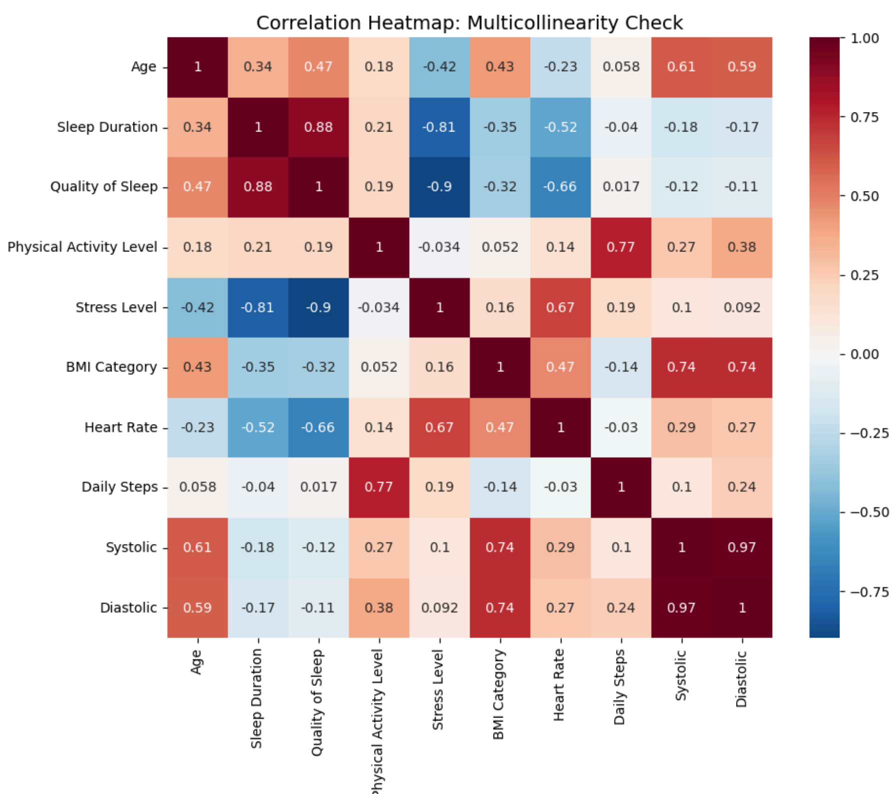

Figure 4: Correlation Heatmap of Numeric Health Indicators.

BMI a Structural Risk Factor

We analyzed the prevalence of sleep disorders across BMI categories to determine if weight was a primary "red flag" (Direction B). The visualization revealed a dramatic "threshold effect":

- Normal Weight individuals rarely exhibited any sleep disorder at all.
- Meanwhile, the Overweight and Obese categories showed a massive surge in Sleep Apnea cases.
- Takeaway: BMI acts as a nearly perfect separator for Sleep Apnea. This visual evidence suggested that a tree-based model also justified the creation of a Random Forest model, which uses binary splits (e.g., Is BMI > Normal?), that would likely excel at classifying the "Sleep Apnea" group.

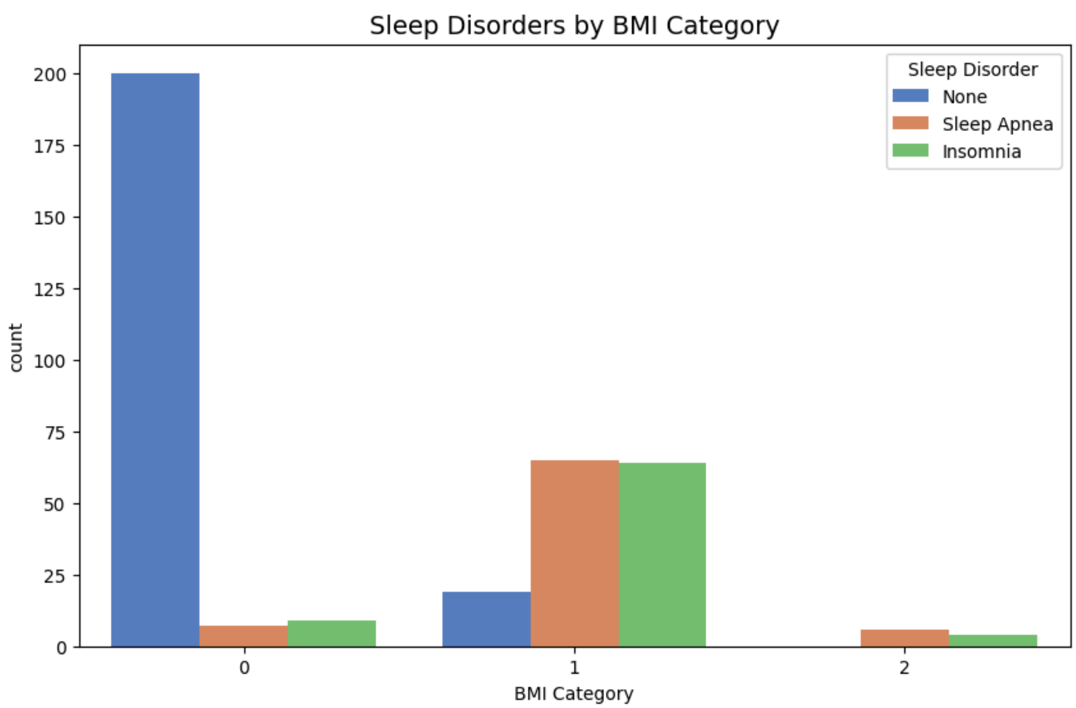

Figure 5: Sleep Disorder Prevalence by BMI Category.

The "Ceiling Effect" of Sleep Quality

Using a Box and Whisker plot to further examine the relationship between Stress Level and Quality of Sleep, we observed an obvious downward pattern. However, a deeper look at the categorical labels revealed that even at low stress levels (levels 3–4), individuals in the Sleep Apnea and Insomnia groups rarely reached the peak sleep quality scores (9–10) enjoyed by the "None" group.

- Takeaway: This revealed that a sleep disorder tends to impose a "ceiling" that lifestyle adjustments (like lowering stress) cannot always overcome. This separation in the "health profiles" provided a strong justification for our K-Nearest Neighbors (KNN) approach, as patients with similar disorders clustered into distinct "neighborhoods" on the Stress-Quality axis.

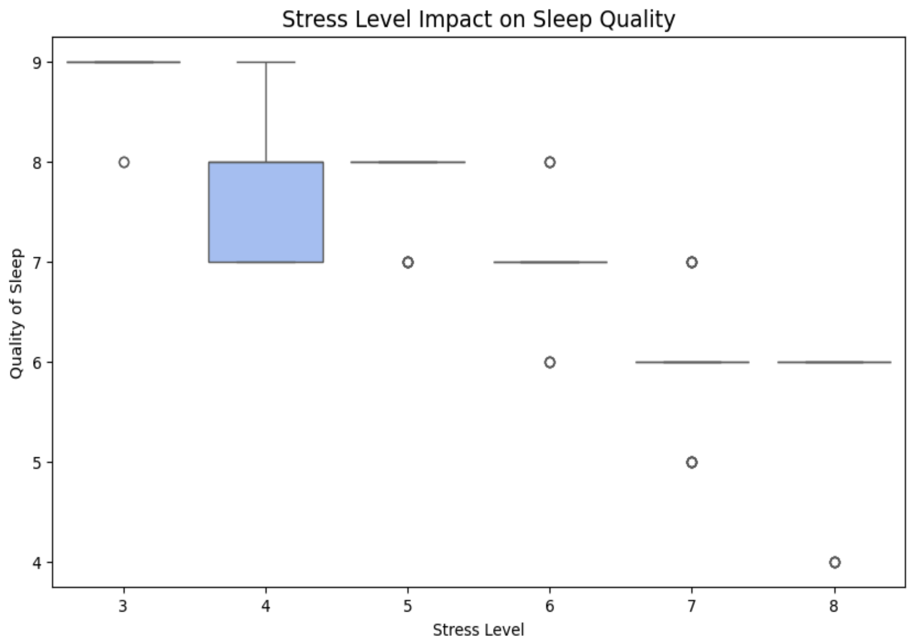

Figure 6: Sleep Quality Distribution across Stress Levels, Segmented by Disorder.

Methodology

To address both Direction A (Prediction) and Direction B (Inference), three distinct algorithms were implemented. Each was chosen for its unique structural approach to classification and interpretability.

Logistic Regression: The Interpretability Baseline

As a parametric model, Logistic Regression was used to establish a baseline for Direction B. By calculating the coefficients, we were able to quantify the "Odds Ratio" of specific health metrics.

- Implementation: We used a multinomial approach to handle the three-class target. While limited by its assumption of linearity, it provided the clearest "clinical" view of how variables like Stress Level directly correlate with disorder risk.

K-Nearest Neighbors (KNN): Neighborhood-Based Profiling

KNN was selected based on the hypothesis that patients with similar lifestyle profiles (e.g., similar age, BMI, and activity levels) would likely share the same sleep disorders.

- Optimization: We utilized 10-fold Cross-Validation to find the optimal K. As shown in our tuning results, K=11 provided the best balance between bias and variance, avoiding the "noise" of small K values while maintaining enough sensitivity to distinguish between Sleep Insomnia and Sleep Apnea.

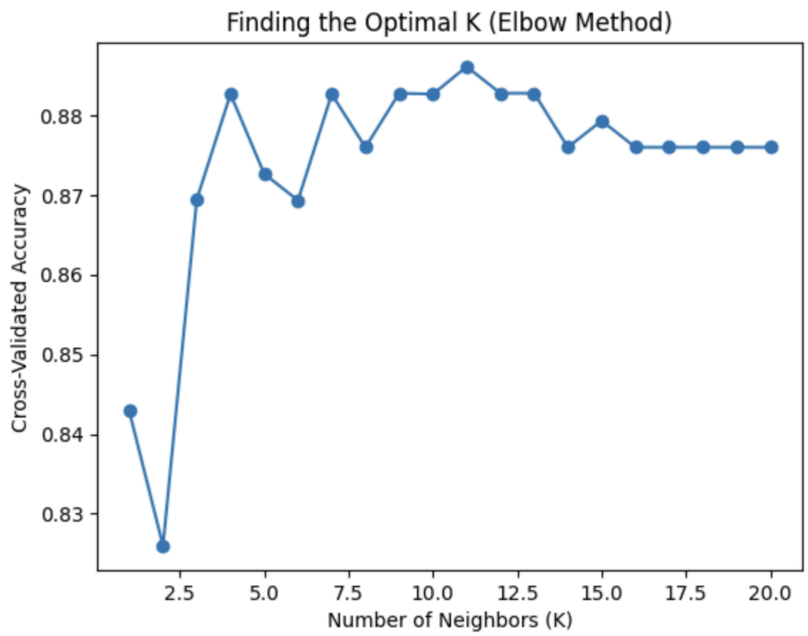

Figure 7: Finding the optimal K Neighbors

Random Forest: The Ensemble Powerhouse

To maximize our predictive accuracy (Direction A), we implemented a Random Forest classifier. By aggregating 50 decision trees, the model was able to capture non-linear interactions, such as the complex relationship between Occupation and Sleep Quality, that simpler models might miss.

- Feature Importance: Unlike the "Black Box" nature of some neural networks, Random Forest allowed us to extract feature importance scores, directly supporting our Inference goals.

Results and Systematic Comparison

Initial Predictive Method Results

A systematic comparison was conducted across the three prediction models: K-Nearest Neighbors (KNN), Logistic Regression, and Random Forest. The models were evaluated using multiple performance metrics, including accuracy, F1-score, AUC-ROC, false positives, false negatives, and cross-validation performance. In addition to predictive performance, interpretability, stability, and computational complexity were also considered.

Based on the initial train-test split evaluation, Random Forest produced the strongest overall performance. It achieved the highest accuracy at 97.3% and the highest AUC-ROC score of 0.998. It also had the fewest classification errors, with zero false positives and zero false negatives in the final evaluation. This suggests that Random Forest was the most effective model at capturing complex relationships and interactions among variables.

Logistic Regression performed nearly as well, with an accuracy of 96.0% and an AUC-ROC score of 0.992. While its predictive performance was slightly lower than Random Forest, it offered a major advantage in interpretability. The model's coefficients and odds ratios made it possible to clearly explain how variables such as stress level, blood pressure, and heart rate affected the likelihood of each sleep disorder.

K-Nearest Neighbors had the lowest overall predictive performance, with an accuracy of 91.0% and an AUC-ROC score of 0.972. It also produced more false positives and false negatives than the other models. However, KNN had the lowest standard deviation during cross-validation, indicating that it was the most stable model across folds, even if it was less accurate overall.

Systemic Comparison with Cross-Validation

After performing 5-fold cross-validation, the comparison shifted slightly. While Random Forest performed best on the initial testing set, Logistic Regression achieved the highest average cross-validation accuracy, suggesting it may generalize better to unseen data. This highlights the importance of evaluating models beyond a single train-test split.

Overall, Random Forest was the strongest model for pure prediction due to its high accuracy and low error rate. However, Logistic Regression provided the best balance between predictive performance and interpretability, making it a strong choice for real-world healthcare applications where transparency is important.

Discussion and Recommendations

Inference Findings (Direction B)

Our analysis confirms that BMI Category and Stress Level are the most significant "red flags" for sleep disorders. Specifically, BMI serves as a primary indicator for Sleep Apnea, while Stress Level is more closely associated with Insomnia. Interestingly, Physical Activity Level showed a weaker direct link, suggesting that exercise alone cannot counteract the negative sleep impacts of high-stress occupations or high BMI.

Final Recommendation

For a real-world health application, we recommend this strategy:

1. For Screening (Logistic Regression): Use the Logistic Regression model for the initial digital screening tool due to its high accuracy across multiple different samples and statistical reliability.
2. For Consultation (Logistic Regression): Use the coefficients from the Logistic Regression to explain *why* a patient is at risk, as clinical users require the "human-readable" logic that ensemble models often obscure.

Conclusion

This project successfully demonstrated that non-invasive lifestyle metrics can predict sleep disorders with high accuracy. By shifting from reactive clinical testing to proactive data-driven screening, healthcare providers can identify high-risk individuals earlier, potentially reducing the long-term cardiovascular risks associated with untreated Sleep Apnea and Insomnia.